

# Intelligence artificielle

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Regression Models: Find a Hidden Function



- 1 Introduction
- 2 Linear Regression
- 3 Non Linear Regression (Bivariate Case)



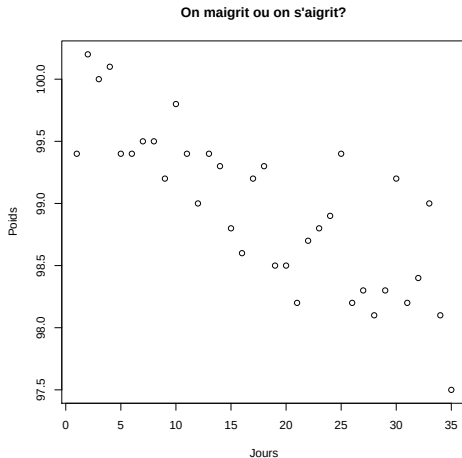
- 1 Introduction
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Let :

- Two variables  $x$  and  $y$ .
- And  $n$  observations of these variables  $(x_1, y_1), \dots, (x_n, y_n)$ .
- We search for the function  $f$  such  $y = f(x)$ .





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- 1 Introduction
- 2 Linear Regression
  - Bivariate Case
  - Multivariate Case
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- 1 Introduction
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    - Introduction
    - Model Evaluation
  - Multivariate Case
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Let :

- Two real variables  $x \in \mathbb{R}$  and  $y \in \mathbb{R}$ .
- And  $n$  observations of these variables  $(x_1, y_1), \dots, (x_n, y_n)$ .
- We know that  $y = f(x)$ .
- We suppose that  $f(x) = bx + a$ .
- We have to find the parameters  $a$  and  $b$ .



To find the most adapted line  $y = bx + a$  to the set of points, we define the residuals ( $\forall i \in \{1, \dots, n\}$ ):

$$e_i = y_i - a - bx_i.$$

If we note  $\hat{y}_i = a + bx_i$  the estimation of  $y_i$ , then

$$e_i = \hat{y}_i - y_i.$$

To find the function that minimizes the residual, we search for the values  $a$  and  $b$  which minimize the sum of squares of the residuals:

$$M(a, b) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (\hat{y}_i - y_i)^2 = \sum_{i=1}^n (y_i - a - bx_i)^2.$$

It is called the least squares method.



It can be shown that the solution of the least squares problem in two dimensions is given by

$$b = \frac{s_{xy}}{s_x^2} \text{ et } a = \bar{y} - b\bar{x}$$

where

- $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$  (mean of  $x$ ).
- $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$  (mean of  $y$ ).
- $s_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$  (variance of  $x$ ).
- $s_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$  (covariance of  $x$  and  $y$ ).

The idea of the linear model is such

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

and some assumptions are mandatory to ensure the good working of this model:

- 1  $E(\varepsilon_i) = 0, \forall i \in \{1, \dots, n\},$
- 2  $E(\varepsilon_i \varepsilon_j) = 0, \forall i \neq j$
- 3  $E(\varepsilon_i^2) = \sigma_\varepsilon^2, \forall i \in \{1, \dots, n\}$

The idea is that  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon)$ .

Assumption 3 is called **homoscedasticity**.

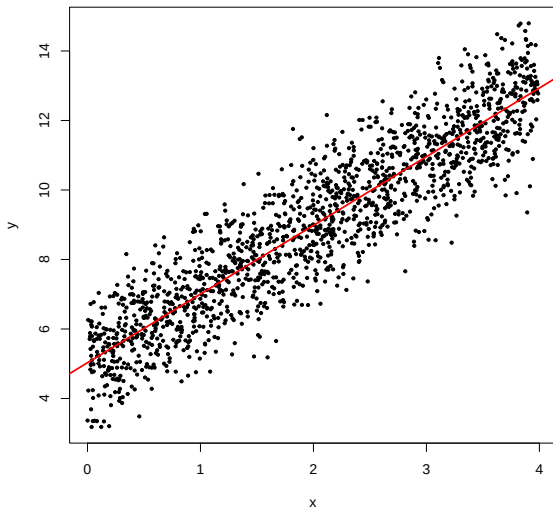
Then the evaluation of the model can be made checking if  $\varepsilon$  follows a normal distribution with a zero mean.



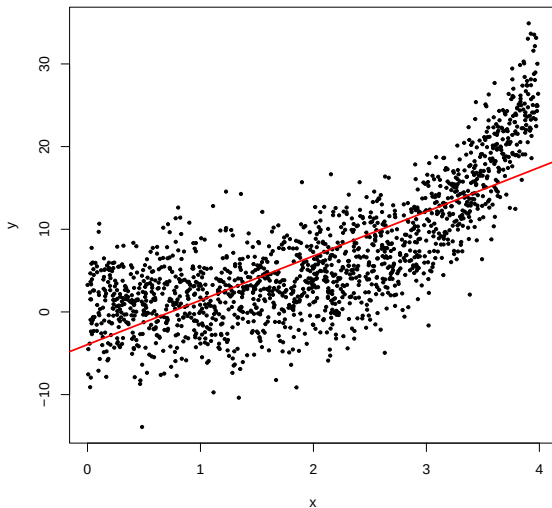
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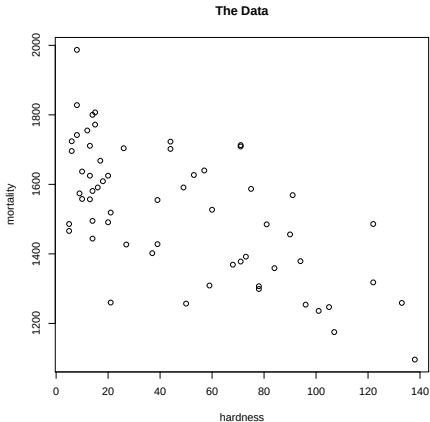
Data are linear



Data are not linear



The idea is to plot the points  $(\hat{y}_i, \varepsilon_i)$  (remember that points  $(x_i, \hat{y}_i)$  are on a line).



The residuals vs fitted plot is useful for checking :

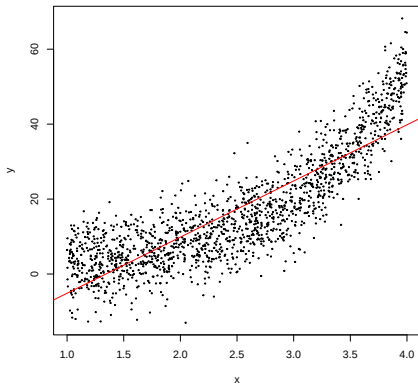
- 1 Whether linearity holds: using the mean residual value line (in red in R) compared to line 0 (dashed line) .
- 2 Whether homoscedasticity holds: the spread of residuals should be approximately the same across the x-axis.
- 3 Whether there are outliers.

Slide inspired by this [blog post](#).



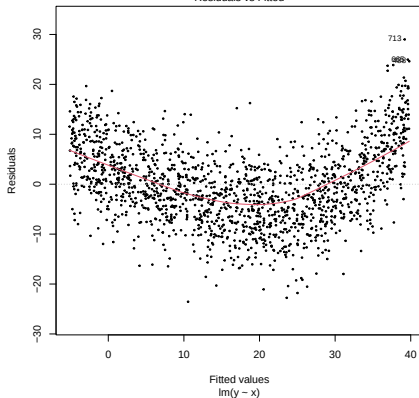
Exponential data with small noise.

Exponential data with small noise



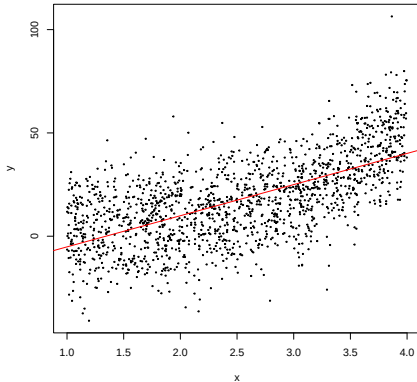
Exponential data with small noise

Residuals vs Fitted



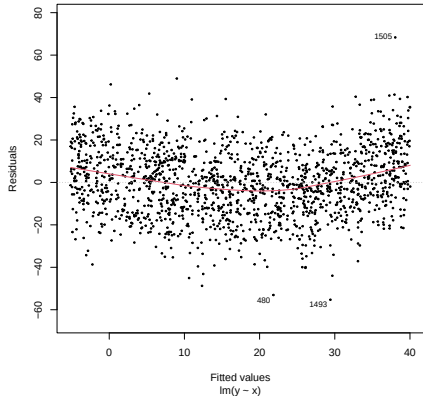
Exponential data with larger noise.

Exponential data with large noise



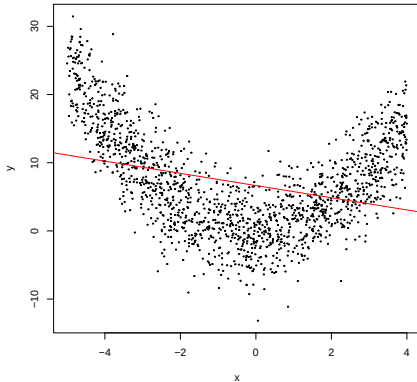
Exponential data with large noise

Residuals vs Fitted



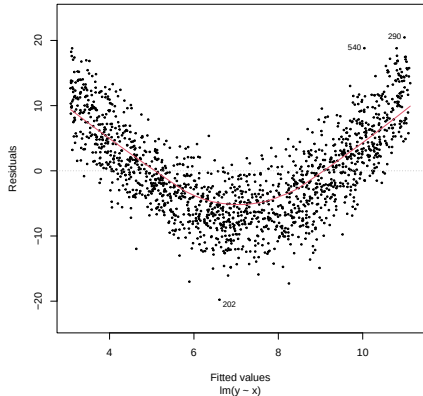
Squared data with small noise.

Squared data with small noise

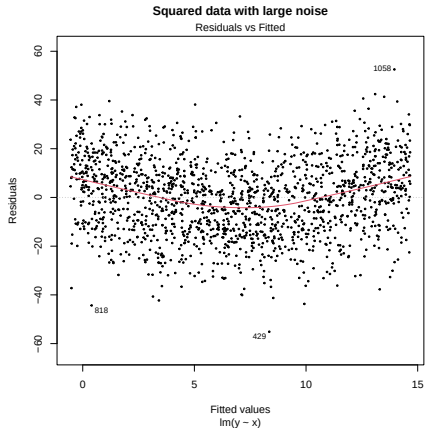
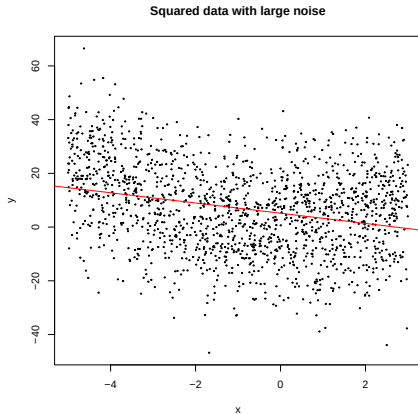


Squared data with small noise

Residuals vs Fitted



Squared data with larger noise.



The idea is to check that  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon)$ , for that, for each  $x_i$  compute :

- the empirical observed distribution:

$$F_{Obs}(e_i) = \frac{|\{e_k \leq e_i\}|}{n} = p_i$$

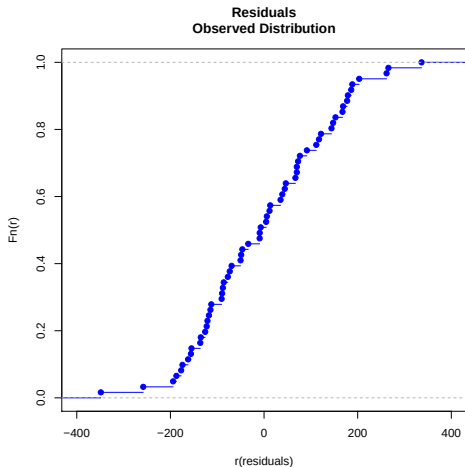
- the theoretical quantile  $\mathcal{N}(0, \sigma_\varepsilon)$ :

$$F_{\mathcal{N}(0, \sigma_\varepsilon)}^{-1}(p_i) = q_i.$$

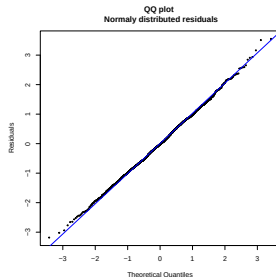
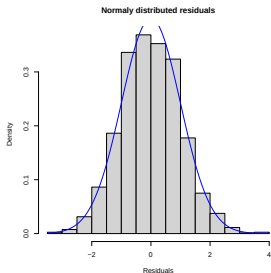
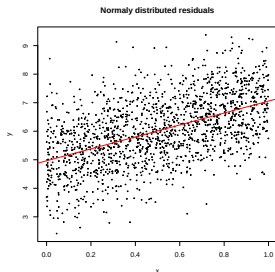
Plot the points  $(q_i, e_i)$ , if  $F_{Obs} = F_{\mathcal{N}(0, \sigma_\varepsilon)}$  then  $\forall i$  we will have  $q_i = e_i$  and then these points will be on the main diagonal.



# Q-Q plot : are the residuals normal ?

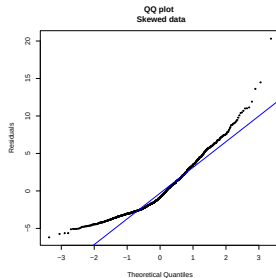
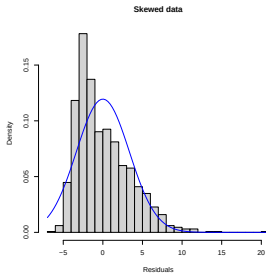
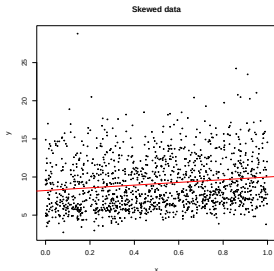


## Normal case.



Slide inspired by this blog post.

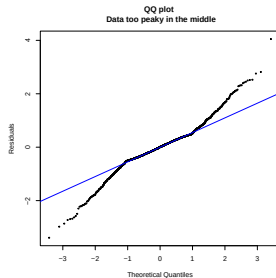
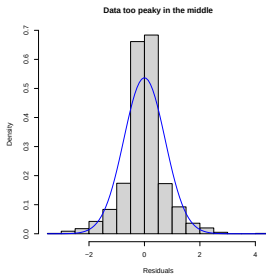
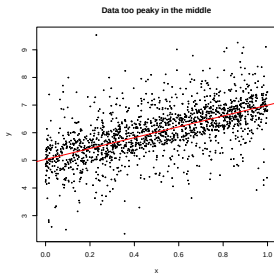
Data skewed<sup>1</sup>.



Slide inspired by this blog post.

<sup>1</sup>Skewness.

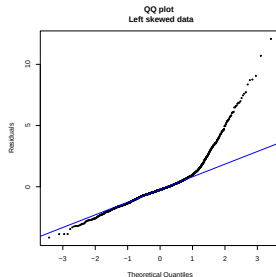
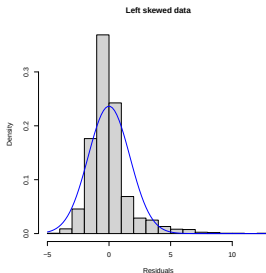
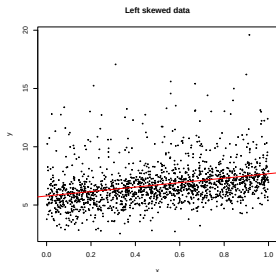
Data too peaky in the middle<sup>2</sup>.



Slide inspired by this blog post.

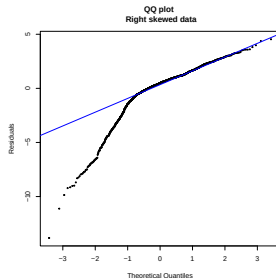
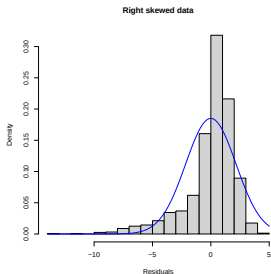
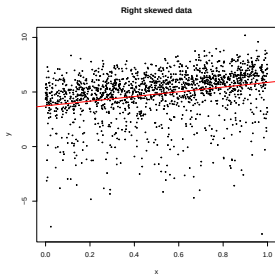
<sup>2</sup>Kurtosis.

## Left skewed residuals.



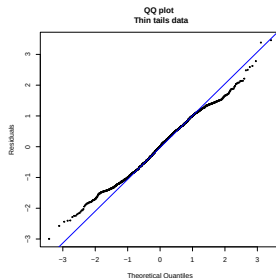
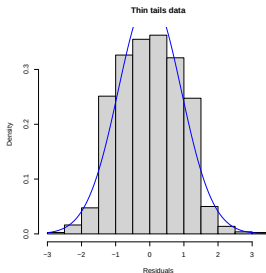
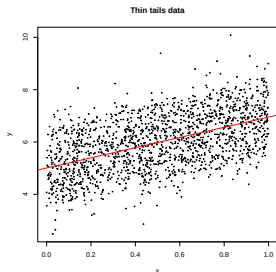
Slide inspired by this blog post.

## Right skewed residuals.



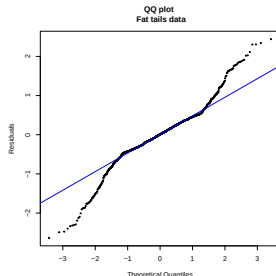
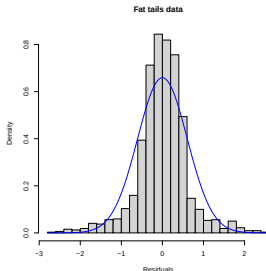
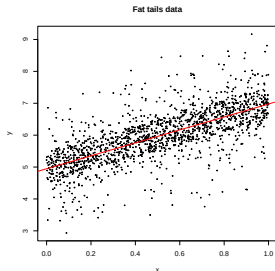
Slide inspired by this blog post.

# Q-Q plots explained : Thin tails residuals



Slide inspired by this blog post.

# Q-Q plots explained : Fat tails residuals



Slide inspired by this blog post.

Total sum of square (proportional to variance):

$$SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2,$$

The explained sum of squares (proportional to explained variance):

$$SS_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2,$$

The residual sum of squares:

$$S_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2 = \sum_i e_i^2$$

## Theorem

$$SS_{\text{tot}} = SS_{\text{reg}} + S_{\text{res}}$$

## Definition

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}} = 1 - \frac{S_{\text{res}}}{SS_{\text{tot}}}$$

- $R^2$  is the proportion of the variation in the dependent variable that is predictable from the independent variable.
- $0 \leq R^2 \leq 1$
- $R^2 \approx 0$  : weak predictive power for the model,
- $R^2 \approx 1$  : strong predictive power for the model.



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Let :

- Two variables  $x \in \mathbb{R}^p$  and  $y \in \mathbb{R}$ .
- And  $n$  observations of these variables  
 $(x_{1,1}, \dots, x_{1,p}, y_1), \dots, (x_{n,1}, \dots, x_{n,p}, y_n)$ .
- We know that  $y = f(x_1, \dots, x_p)$ .
- We suppose that

$$f(x_1, \dots, x_p) = b_0 + b_1x_1 + \dots + b_px_p.$$

- We have to find the parameters  $(b_0, \dots, b_p)$ .



The observations  $(x_{1,1}, \dots, x_{1,p}, y_1), \dots, (x_{n,1}, \dots, x_{n,p}, y_n)$  and the parameters  $(b_0, \dots, b_p)$  can be noted

$$\mathbf{X} = \begin{pmatrix} 1 & x_{1,1} & \cdots & x_{1,p} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n,1} & \cdots & x_{n,p} \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} y_1 \\ \vdots \\ x_n \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_0 \\ \vdots \\ b_p \end{pmatrix}$$

If we remark that

$$f(x_{j,1}, \dots, x_{j,p}) = b_0 1 + b_1 x_{j,1} + \cdots + b_p x_{j,p}.$$

then the searched parameters are such

$$\mathbf{Y} = \mathbf{Xb}$$



The searched parameters  $(b_0, \dots, b_p)$  are those which minimizes the sum of squared residuals:

$$\begin{aligned} Q(b_0, \dots, b_p) &= \sum_{i=1}^n e_i^2 \\ &= \sum_{i=1}^n [y_i - (b_0 + b_1 x_{i,1} + \dots + b_p x_{i,p})]^2 \\ &= \sum_{i=1}^n \left( y_i - \sum_{j=0}^p b_j x_{i,j} \right)^2 \end{aligned}$$

If we remark that

$$\mathbf{e} = \mathbf{Y} - \mathbf{bX} = \begin{pmatrix} y_1 - \sum_{j=0}^p b_j x_{1,j} \\ \vdots \\ y_n - \sum_{j=0}^p b_j x_{n,j} \end{pmatrix}$$

Then the function  $Q$  can be rewritten

$$Q(b_0, \dots, b_p) = Q(\mathbf{b}) = \sum_{i=1}^n e_i^2 = \mathbf{e}^T \mathbf{e} = (\mathbf{Y} - \mathbf{bX})^T (\mathbf{Y} - \mathbf{bX}) = \|\mathbf{Y} - \mathbf{bX}\|^2$$

it can be shown the solution is given by

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

if the inverse of  $\mathbf{X}^T \mathbf{X}$  exists.



The idea of the linear model is such

$$y_i = \alpha + \sum_{j=0}^p x_{i,j} \beta_j + \varepsilon_i$$

and some assumptions are mandatory to ensure the good working of this model:

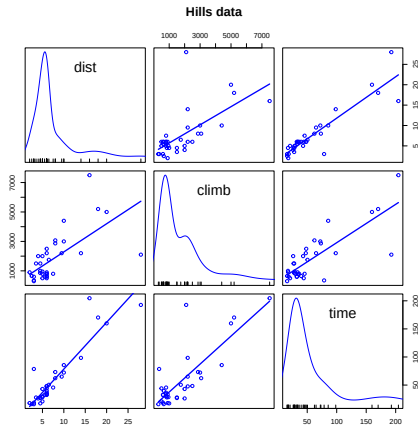
- $E(\varepsilon_i) = 0, \forall i \in \{1, \dots, n\},$
- $E(\varepsilon_i \varepsilon_j) = 0, \forall i \neq j$
- $E(\varepsilon_i^2) = \sigma_\varepsilon^2, \forall i \in \{1, \dots, n\}$

The idea is that  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon)$ .

Then the evaluation of the model can be made checking if  $\varepsilon$  follows a normal distribution with a zero mean.

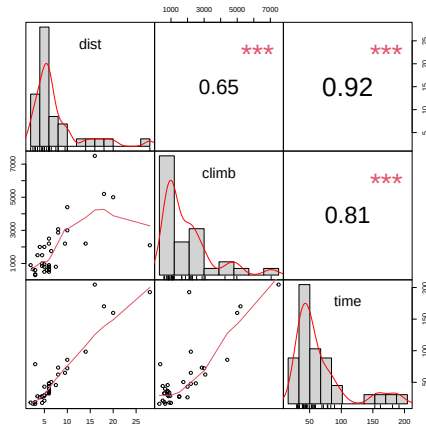
The same tools than in the bivariate case can be used.





Regression lines in this graph are not the projections of the multivariate regression...

... because correlation between variable is not taken into account.



Let

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

where,  $Y$  is the dependant variable,  $X_2$ ,  $X_3$  the independant variable and  $\epsilon$  a normal noise.

Suppose we omit  $X_3$  from the regression, and suppose the relation between  $X_2$  and  $X_3$  is given by

$$X_3 = \gamma_1 + \gamma_2 X_2 + e$$

where  $e$  is also a normal noise.

Substituting the second equation into the first gives

$$\begin{aligned} Y &= \beta_1 + \beta_2 X_2 + \beta_3 (\gamma_1 + \gamma_2 X_2 + e) + \epsilon \\ &= (\beta_1 + \beta_3 \gamma_1) + (\beta_2 + \beta_3 \gamma_2) X_2 + (\beta_3 e + \epsilon) \end{aligned}$$

This last equation is what is estimated if a regression of  $Y$  is conducted upon  $X_2$  only, and most of the times  $\beta_2 + \beta_3 \gamma_2 \neq \beta_2$ .

In the regression for

$$Y = \beta_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

it is important to interpret coefficients  $\beta_2$  and  $\beta_3$  by considering what happens when the other variables are held constant, which is not the case in the equation

$$Y = (\beta_1 + \beta_3 \gamma_1) + (\beta_2 + \beta_3 \gamma_2) X_2 + (\beta_3 e + \epsilon)$$

For the sake of illustration, suppose that  $\beta_2, \beta_3 > 0$  while  $\gamma_2 < 0$ . In the first equation  $Y$  increase when  $X_2$  increase, while in the second case, the effect of  $X_2$  via  $X_3$  could make in some case that  $Y$  decrease when  $X_2$  increase.



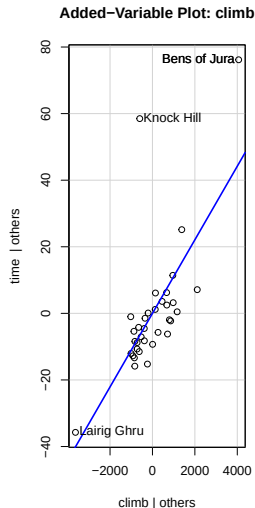
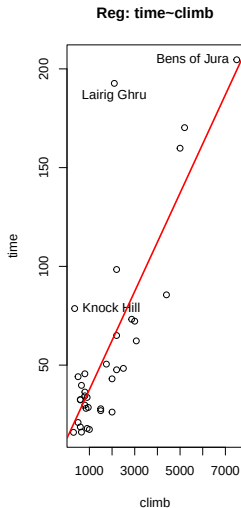
For our example, the idea is to :

- 1 regress  $X_2$  upon  $X_3$  and use the residuals from this regression, they are the values of  $X_2$  purged of the influence of  $X_3$ ,
- 2 regress  $Y$  upon  $X_3$  and use the residuals from this regression, they are the values of  $Y$  purged of the influence of  $X_3$ ,
- 3 plot values from (1) values against values from (2) .

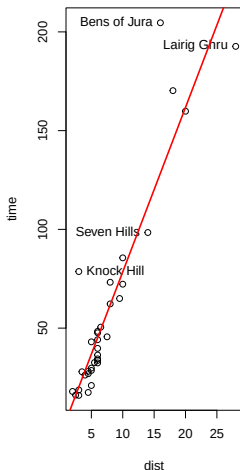
In general, for each variable  $X_i$  we

- 1 compute the residuals from regressing  $X_i$  against the remaining independent variables,
- 2 compute the residuals of regressing the response variable against the independent variables but omitting  $X_i$ ,
- 3 plot the residuals from (1) against the residuals from (2).

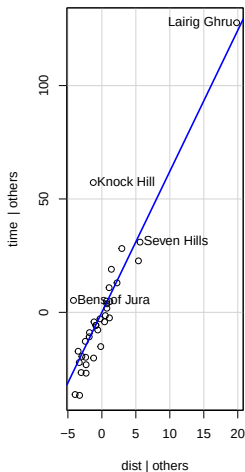




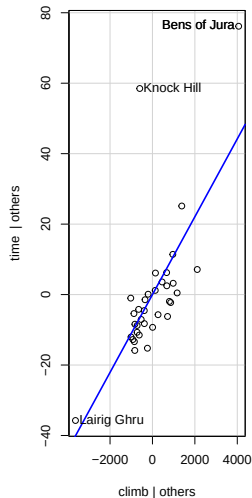
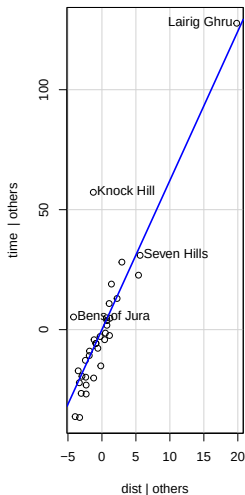
Reg: time~dist



Added-Variable Plot: dist



Added-Variable Plots



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  - Parametric estimations
  - Non parametric estimation (LOESS)



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- We know that  $y = f(x)$ .



The idea of the non linear model is such

$$y_i = f(x_i; \beta) + \varepsilon_i$$

and some assumptions are mandatory to ensure the good working of this model:

- $E(\varepsilon_i) = 0, \forall i \in \{1, \dots, n\},$
- $E(\varepsilon_i \varepsilon_j) = 0, \forall i \neq j$
- $E(\varepsilon_i^2) = \sigma_\varepsilon^2, \forall i \in \{1, \dots, n\}$

The idea is that  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon)$ .

Then the evaluation of the model can be made checking if  $\varepsilon$  follows a normal distribution with a zero mean.



- Exponential:

$$y_i = \alpha * \exp(x_i * \beta) + \varepsilon_i$$

- Polynomial:

$$y_i = a_0 + a_1 * x_i^1 + \dots + a_n * x_i^n + \varepsilon_i$$

- Splines functions (cf. next slide)

- ...



A general form of non-linear regression is

$$\hat{f}(t) = \sum_{i=1}^K c_i \phi_i(t)$$

where  $\phi_i$  are the basis functions.

The most familiar basis functions are the monomials:

$$1, t, t^2, t^3, \dots, t^k, \dots$$

or the Fourier series terms

$$1, \sin(\omega t), \cos(\omega t), \sin(2\omega t), \cos(2\omega t), \dots$$

The most flexible are based on spline functions, and more precisely on B-splines.



A spline is a function defined piecewise by polynomials: let  $K$  points, called knots, belonging to a real interval  $[a, b]$ :

$a = t_0 < t_1 < \dots < t_{K-2} < t_{K-1} = b$ . The function

$$S : [a, b] \rightarrow \mathbb{R}$$

is called a spline of degree  $n$  if

- 1  $S$  is  $n - 1$  differentiable with continuous differentials,
- 2 for any restriction on  $[t_i, t_{i+1}]$ ,  $S$  is a polynomial of degree  $n$ .

Thus:

- a spline of degree 0, is just a piecewise constant function, and its derivative is the zero function,
- a spline of degree 1, is just a piecewise linear function, and its derivative is a piecewise constant function,
- a spline of degree 2, is just a piecewise quadratic function, and its derivative is a piecewise linear function,
- ...



The most powerful splines functions are the B-spline functions: given the knots  $\tau = \{a = t_0, t_1, \dots, t_{K-2}, t_{K-1} = b\}$  defined above, a B-spline of degree  $n$  is given by the following linear combination:

$$S(t) = \sum_{k=0}^{K+n-1} c_k B_{k,n}(t; \tau)$$

where  $n$  is the degree of the basis functions  $B_{k,n}$ . These latter are defined as follow:

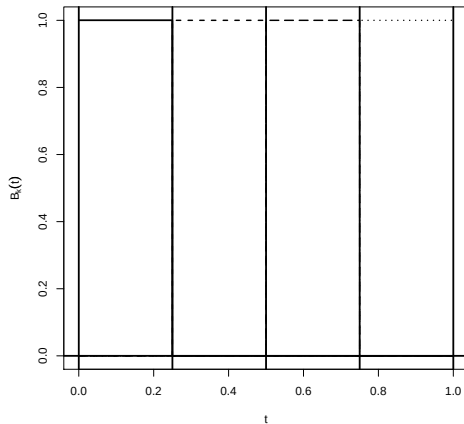
$$B_{j,0}(t; \tau) = \begin{cases} 1 & \text{if } t_j \leq t < t_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

$$B_{j,n}(t; \tau) = \frac{t - t_j}{t_{j+n} - t_j} B_{j,n-1}(t; \tau) + \frac{t_{j+n+1} - t}{t_{j+n+1} - t_{j+1}} B_{j+1,n-1}(t; \tau)$$

where  $B_{k,n}(t; \tau)$  denotes the value at  $t$  of the  $k^{\text{th}}$  function of the basis defined by the set of knots  $\tau$ .



Basis functions of a B-spline of order 0



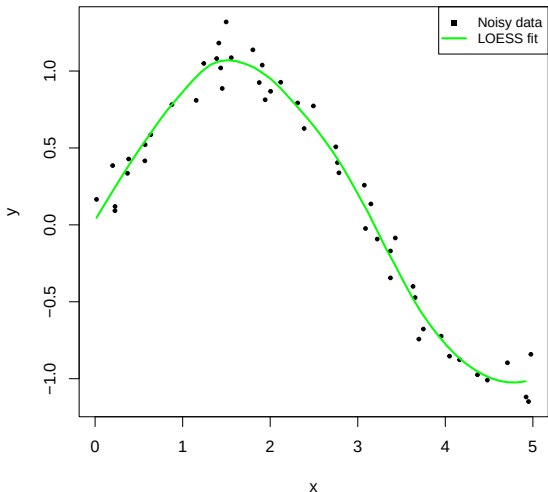
- 1 Introduction
- 2 Linear Regression
- 3 Non Linear Regression (Bivariate Case)
  - Parametric estimations
  - Non parametric estimation (LOESS)



- LOESS stands for Locally Estimated Scatterplot Smoothing
- It is a non-parametric regression technique
- Fits a smooth curve to data without assuming a global model
- Builds local models around each point using nearby data



LOESS smoothing



- Ideal for exploratory data analysis (EDA)
- Captures non-linear patterns without specifying a functional form
- Useful for visualizing trends in noisy datasets
- Reveals local variations missed by global models



- For each target point:
  - Select nearby points (based on span)
  - Assign weights (closer points have higher weight)
  - Fit a polynomial regression (linear or quadratic)
- Repeat for all points to create a smooth curve



Given  $S = \{(x_1, y_1), \dots, (x_N, y_N)\} \subset [a, b] \times [c, d]$  with  $a, b, c, d \in \mathbb{R}$ .  
For each point  $x$  chosen in  $[a, b]$ , LOESS minimizes:

$$\sum_{i=1}^n K_i \cdot (y_i - \hat{y}_i)^2$$

Where:

- $K_i = K(x_i; x)$ : weight of point  $x_i$  based on distance to  $x$ ,
- $\hat{y}_i = \hat{f}(x_i; \bar{\beta})$ : predicted value from local polynomial fit,
- $\bar{\beta}$  parameters for  $\hat{f}$  computed for  $x$ ,
- $n = |V(x)|$  where  $V(x)$  is the neighbourhood of  $x$  in  $S$ .



## Linear Case (Degree = 1):

$$\sum_{i=1}^n K_i \cdot (y_i - (\beta_0 + \beta_1 x_i))^2$$

$$\hat{f}(x) = \beta_0 + \beta_1 x$$

## Quadratic Case (Degree = 2):

$$\sum_{i=1}^n K_i (y_i - (\beta_0 + \beta_1 x_i + \beta_2 x_i^2))^2$$

$$\hat{f}(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$



- **Span** ( $s$ ): proportion of data used for local fitting
- **Degree**: 1 = linear, 2 = quadratic
- **Weight function**:

$$K(x_i; x) = w\left(\frac{|x - x_i|}{D_x}\right)$$

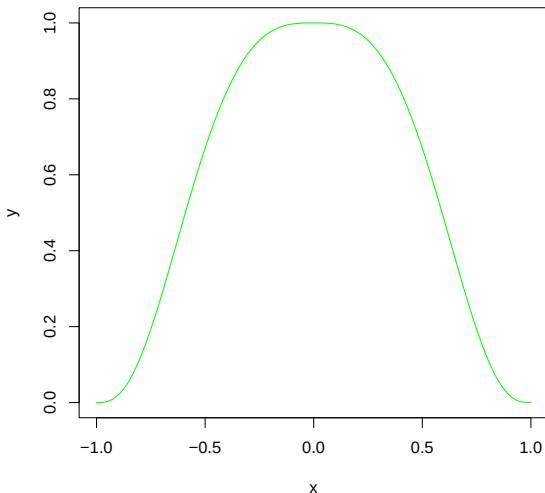
with  $D_x = \max(\{|x - x_k|, x_k \in V(x)\})$

- Typically  $w$  is the tri-cube weighting

$$w(x) = (1 - |x|^3)^3$$



Tri cube function



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## Algorithm 1 Local regression at $x \in [a, b]$

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**Input:**  $S = \{(x_1, y_1), \dots, (x_N, y_N)\} \subset [a, b] \times [c, d]$  with  $a, b, c, d \in R$

**Output:** estimation function  $\hat{f}$

**Parameters:**  $s$

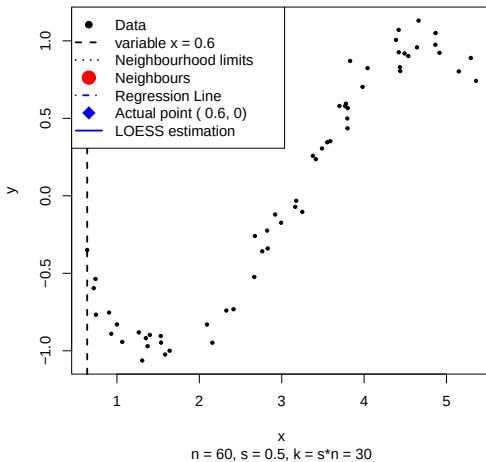
- 1: **for** each  $x \in [a, b]$  **do**
- 2:   Gather in  $S$  the  $k$  neighbours of  $x$  with  $s = k/N$ , call it neighbourhood  $V(x)$ .
- 3:   For each  $x_i \in V(x)$  compute a weight  $K_i = K(x_i; x)$ .
- 4:   Find  $\beta_0$  and  $\beta_1$  using a weighted least square regression on  $\{(x_i, y_i) \in V(x)\}$ , with  $n = |V(x)|$ :

$$\sum_{i=1}^n K_i \cdot \left( y_i - \hat{f}(x_i; \beta_0, \beta_1) \right)^2.$$

- 5:   The fitted value at  $x$  is given by  $\hat{f}(x; \beta_0, \beta_1)$
- 6: **end for**



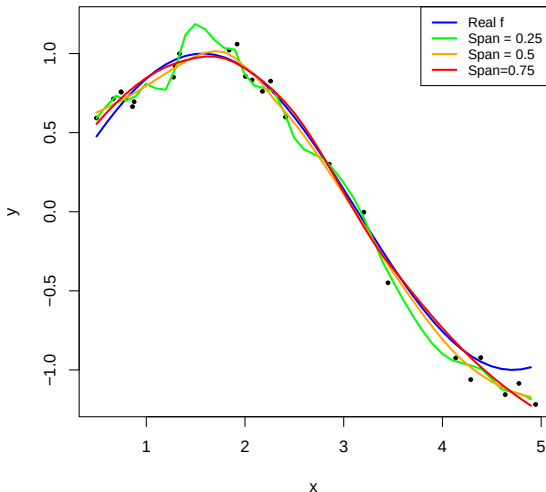
## LOESS smoothing



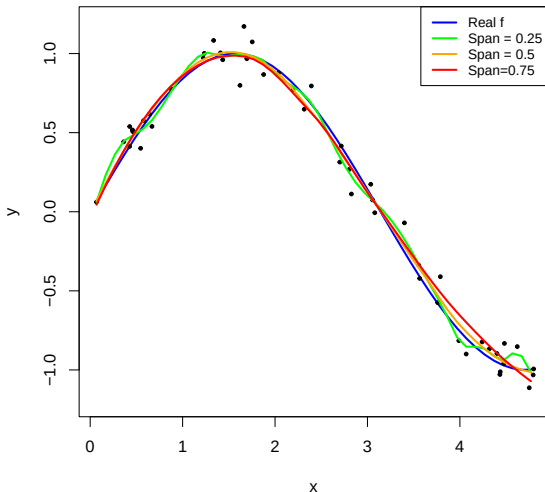
- Scatterplot of noisy data
- LOESS curve adapts to local structure
- Compare with linear regression: LOESS follows data more closely

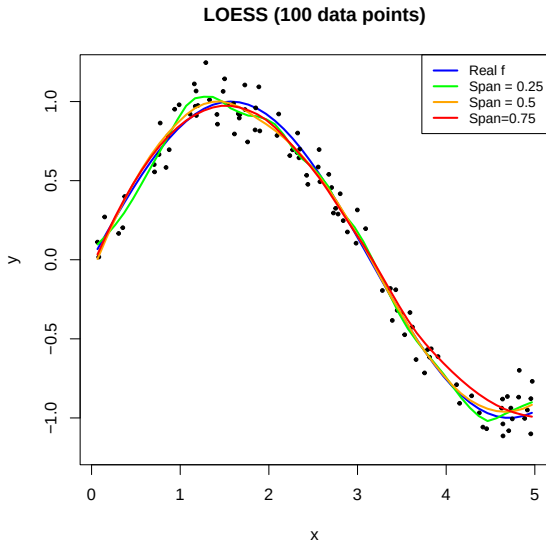


LOESS (25 data points)



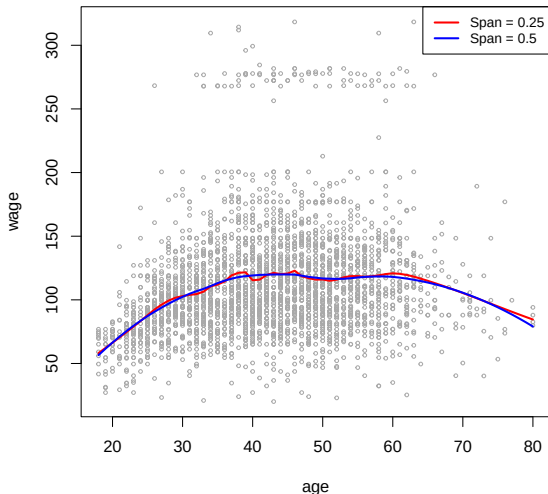
LOESS (50 data points)





| Method                | Assumes Model | Non-linearity | Local Fit | Interpretability |
|-----------------------|---------------|---------------|-----------|------------------|
| Linear Regression     | Yes           | Poor          | No        | High             |
| Polynomial Regression | Yes           | Moderate      | No        | Moderate         |
| LOESS                 | No            | Excellent     | Yes       | Low              |
| Spline                | Yes           | Excellent     | Yes       | Moderate         |

LOESS on real data



- LOESS is a powerful non-parametric smoothing technique
- Builds local models to capture complex patterns
- Best used for EDA and visualization
- Helps reveal the story in your data

